

Safety Helmet Detection: Technical Report

1. Executive Summary

The **Safety Helmet Detection** system is a specialized computer vision solution designed to enhance industrial safety by monitoring compliance with hard hat regulations. Utilizing the YOLO (You Only Look Once) architecture and optimized for edge deployment via TensorFlow Lite (TFLite), the system provides real-time detection of personnel and their safety gear. This report outlines the technical framework, model performance, and deployment strategy of the project.

2. Project Objectives

The primary objectives of this project include:

- **Real-Time Compliance Monitoring:** Automatically identifying individuals on-site and determining if they are wearing required safety helmets.
- **Edge-Ready Deployment:** Optimizing the detection model for low-latency performance on resource-constrained devices.
- **Automated Safety Alerts:** Providing a visual interface that highlights non-compliant personnel for immediate corrective action.

3. Technical Architecture

3.1 Object Detection Model

The system leverages the **YOLOv8** architecture, known for its balance of speed and accuracy. The model was trained on the “Hard Hat Detection” dataset from Kaggle, which includes a diverse range of industrial environments and lighting conditions.

3.2 Model Optimization

To ensure the system can be deployed in real-world scenarios (e.g., on mobile devices or embedded cameras), the model was converted to the **TensorFlow Lite (TFLite)** format.

- **Format:** `best_float32.tflite`
- **Optimization:** Float32 quantization to maintain high precision while reducing model size and inference time.

3.3 Software Stack

Component	Technology
Detection Framework	Ultralytics YOLO
Inference Engine	TensorFlow Lite
Image Processing	OpenCV
User Interface	Gradio
Language	Python 3.x

4. Detection Classes and Logic

The system is trained to recognize and categorize three distinct classes:

Class	Description	Visual Indicator	Safety Status
Helmet	Personnel wearing a safety helmet.	Green Bounding Box	Compliant
Head	Personnel with a bare head (no helmet).	Red Bounding Box	Non-Compliant
Person	General detection of a human figure.	Blue Bounding Box	N/A

The detection logic applies a confidence threshold of **0.25** to minimize false positives while ensuring high recall in critical safety zones.

5. System Workflow

1. **Image Acquisition:** Images are captured via the Gradio web interface or a connected camera stream.
2. **Preprocessing:** Input images are resized and normalized to match the TFLite model's input requirements.
3. **Inference:** The TFLite engine processes the image and returns bounding box coordinates, class labels, and confidence scores.
4. **Post-processing:** Bounding boxes are filtered by confidence and drawn onto the original image with color-coded indicators.
5. **Output:** The processed image is displayed to the user, providing an immediate visual audit of safety compliance.

6. Conclusion

The Safety Helmet Detection system provides a robust and scalable tool for industrial safety management. By combining the power of YOLOv8 with the efficiency of TFLite, the project delivers a high-performance solution capable of operating in real-time environments. Future developments may include integration with CCTV systems for continuous monitoring and the addition of other safety gear detection (e.g., high-visibility vests, safety goggles).

Report generated for the Safety Helmet Detection Project